

OCA I3C Controller — Test Plan

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Style: SystemC / TLM-2.0 LT · self-checking benches · CCI 1.0

This plan enumerates the verification intent for the OCA I3C Controller model. Two self-checking benches implement it; each prints `ALL TESTS PASSED` on success and exits non-zero on any failure. They are registered with CTest and run by `run_tests.sh`.

1. Strategy

- **Primary bench** (`i3c_controller_tb.cpp`): functional coverage against a running kernel — register access, multi-instance decode, device tables, the full HCI transaction flow (write/read/CCC), error paths, IBI, interrupt aggregation, queue resets, and CCI introspection. Uses a 3-instance DUT and attaches a bus model emulating a single target at dynamic address `0x42`.
 - **Negative bench** (`i3c_controller_neg_tb.cpp`): construction guard rails and TLM/ `transport_dbg` error paths, exercised during elaboration without `sc_start` (so constructor-throw probes do not perturb the kernel).
 - **Instrumentation:** `--asan` (AddressSanitizer, + LSan on Linux) must report zero errors; `--coverage` produces a line-coverage report.
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2. Primary bench cases

#	Area	Intent
1	Reset values	HCI_VERSION=0x120 , HC_CONTROL=0x40 (MODE_SELECTOR=1), QUEUE_THLD_CTRL / DATA_BUFFER_THLD_CTRL=0x01010101 , PIO_CONTROL=0x03 , PIO_SECTION_OFFSET=0x80 , STBY_CR_EXTCAP_HEADER=0x12 , DAT_SECTION_OFFSET fields.
2	RW / RO / W1C	RW round-trip (CONTROLLER_DEVICE_ADDR); RO ignore (HCI_VERSION); INTR_FORCE set + INTR_STATUS W1C clear.
3	DAT / DCT windows	Write/read-back of DAT and DCT direct-access entries.
4	Bus enable	HC_CONTROL.BUS_ENABLE ; PRESENT_STATE.AC_CURRENT_OWn follows.
5	Private write	TX push + command → response (TID/len/SUCCESS); bus model receives little-endian bytes.
6	Private read	Read command → response + RX payload via XFER_DATA_PORT .
7	CCC write	CP=1 , CCC code 0x80 → response SUCCESS; payload delivered.
8	Address NACK	Command to an unpopulated DAT index → ERROR_ADDRESS_NACK .
9	TX underflow	Write len=8 with one TX DWORD → ERROR_OVERFLOW/UNDERFLOW ; TRANSFER_ERR_STAT latched + W1C clear.
10	IBI injection	inject_ibi(0,0x42,{0xAA,0xBB}) → IBI_PORT status (addr<<1 len<<8) + payload DWORD.
11	Interrupt aggregation	Enable RESP_READY signal → irq_o[0] high while a response is pending, low after drain.
12	RESET_CONTROL	CMD_QUEUE_RST clears a queued command; register self-clears (reads 0).
13	Multi-instance isolation	Writes to instance 1 do not affect instance 0/2; per-instance DAT isolation.
14	transport_dbg	Back-door DAT write via socket + dbg_read snapshot match.
15	QUEUE_SIZE	Reflects configured CR (8) and IBI (8) depths.
16	CCI introspection	access_delay_ns mutate; num_instances reads the resolved value.

3. Negative bench cases

#	Area	Intent
1	Constructor guards	<code>num_instances=0</code> , <code>=7</code> , <code>access_delay_ns<0</code> , <code>xfer_delay_ns<0</code> , <code>tx_fifo_depth=0</code> each raise <code>SC_REPORT_FATAL</code> (probed via throwing handler).
2	TLM errors	Non-word width, misaligned, streaming-width mismatch, byte-enable, out-of-aperture, bad command → the matching <code>TLM*_ERROR_RESPONSE</code> .
3	RAZ/WI hole	Unmapped in-window offset reads 0, write accepted.
4	transport_dbg	Misaligned / bad-width / OOB / CSR (non-table) accesses return 0; DAT-window write/read round-trip returns 4.

4. Pass / fail criteria

- Both benches print `ALL TESTS PASSED`; CTest reports 2/2 passing.
- `--asan` reports **0** errors.
- No SystemC `Error` / `Fatal` outside the negative bench's expected probes.

5. Known coverage gaps (by design)

The following are out of scope for the LT model and therefore unverified here (see `01_..._Specification.md §12`): bit-level SCL/SDA waveforms, OD/PP timing parameters, CDC/metastability, HDR modes, the TCRI recovery datapath, the external 128-entry DAT/DCT memory interface, and scan/DFT.